Praat → LuaLaTeX Pipeline

Spectrograms with TikZ-drawn alignment tiers

Your Name

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This worked example shows the pipeline used by phoneticians and computational linguists to publish aligned acoustic data: export a spectrogram from Praat as a PDF, embed it with `graphicx`, and draw the phoneme / word / gloss tiers over it with TikZ.

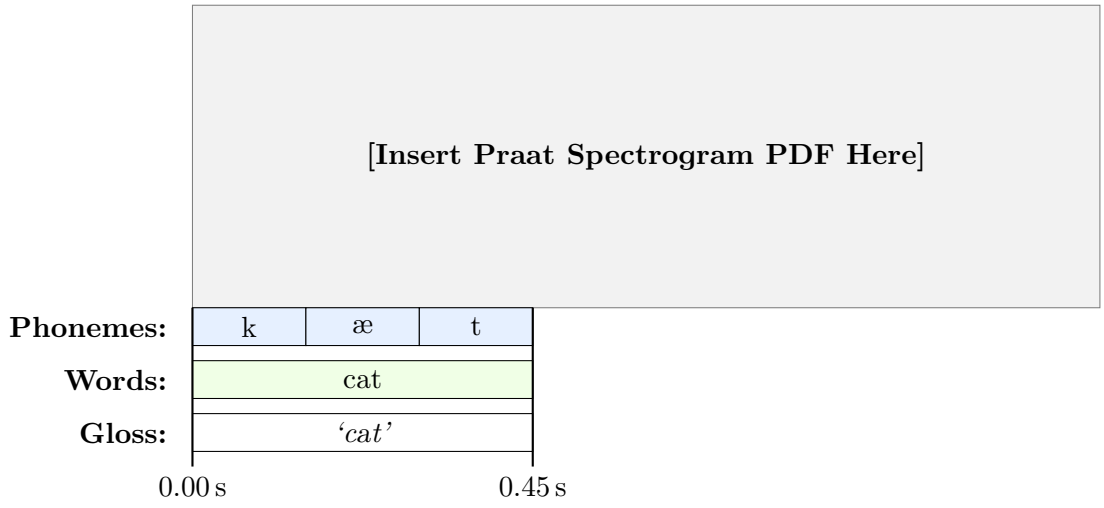
1 Step 1 — Export from Praat

In Praat, select the Sound and the TextGrid together, open **View & Edit**, show the spectrogram and the tiers you want, then **File → Save as PDF...** Call the file `spectrogram.pdf`.

For automated workflows, export the TextGrid as a text or CSV file and read it from Python with `parselmouth` or `textgridtools`. That gives you the exact start/end time of every segment in seconds, which you can feed directly into the TikZ coordinates below.

2 Step 2 — Embed and overlay in LaTeX

The example below uses a gray placeholder box in place of a real spectrogram so this file compiles out of the box. Replace it with `\includegraphics` once you have a real `spectrogram.pdf`.



3 Step 3 — Scaling the coordinates

TikZ draws in abstract units, but the spectrogram sits in seconds. If your audio file is T seconds long and you embedded the spectrogram at width W cm, then an event starting at time t seconds has x-coordinate

$$x = \frac{t}{T} \cdot W \quad \text{(cm)}.$$

Example: a vowel starting at 0.3 s in a 1.5 s file, with a 12 cm wide spectrogram, starts at $x = (0.3/1.5) \times 12 = 2.4$ cm.

For anything more than a handful of segments, let Python do the arithmetic. A short script can read the TextGrid and spit out `\phonbox{...}{...}{...}` lines directly, which you then `\input{}` into the TikZ picture. LaTeX does the drawing; Python does the math.

4 Related packages

- `tikz-dependency` — for dependency parsing: draws labeled arcs over text (useful for spaCy or UDPipe output).
- `expex`, `linguex` — alternatives to `gb4e` for numbered glosses; some departments prefer them.
- `forest` — phrase-structure trees (already in the main kit).